AI-Driven Fish Biometric Estimation for Smart Aquaculture Systems

1st Santiago Crespo López
Faculty of engineering
Universidad Militar Nueva Granada
Bogotá D.C., Colombia
est.santiago.crespo@unimilitar.edu.co

2nd Alexandra Velasco-Vivas
Faculty of engineering
Universidad Militar Nueva Granada
Bogotá D.C., Colombia
alexandra.velasco@unimilitar.edu.co

3rd Leonardo Solaque
Faculty of engineering
Universidad Militar Nueva Granada
Bogotá D.C., Colombia
leonardo.solaque@unimilitar.edu.co

***Abstract*—Traditional practices for measuring fish length and weight in aquaculture involve direct manipulation of the animal and anesthetic chemicals, negatively affecting welfare and growth. This work proposes a non-intrusive stereo-vision system for length measurement and weight estimation, tested on *Piaractus brachypomus*. The methodology integrates stereo image acquisition, camera calibration, fish detection and keypoint localization using YOLO-Fish and HRNet-W32, and post-processing via epipolar geometry for depth estimation and length measurement, followed by weight estimation through a calibrated allometric model. The relative error was 6.7% for length and 5.87% for weight, using low-cost hardware and computationally efficient processing. These results demonstrate the feasibility of integrating stereo vision, deep-learning-based keypoint localization, and allometric weight estimation for non-intrusive fish biometric assessment.**

***Index Terms*—Deep Learning, Stereo Vision, Keypoint Detection, Biometric Measurement, Aquaculture Automation, Non-invasive Monitoring, Allometric Model.**

I. INTRODUCTION

Measurement of fish weight and length is fundamental in modern aquaculture for assessing growth, health, welfare, and production performance [1]. These measurements are also essential for determining feed quantity and optimizing production efficiency [1], [2]. However, traditional methods, used in approximately 85% of commercial production systems [1], require capturing and handling fish with anesthetic agents such as MS-222, introducing human error, additional costs, and inaccuracies in biomass estimation [1], [3], [4]. Moreover, handling substantially increases stress, suppressing appetite, hindering growth, increasing disease susceptibility, and potentially causing mortality [5], [6]. These limitations highlight the need for automated, non-invasive systems that provide accurate biometric data without fish manipulation. Consequently, computer vision has emerged as a cost-effective and animal-friendly solution for efficient and robust monitoring in modern aquaculture [5].

Within the domain of non-intrusive fish detection and measurement, four main technological approaches have been identified: computer vision–based systems for extracting morphological characteristics from captured images, acoustic-based systems employing sound waves for fish detection and counting, environmental DNA (eDNA)-based systems utilizing DNA extraction from water and quantification through PCR techniques, and electrical resistivity counter systems measuring changes in water electrical resistance as fish pass between electrodes [7]. Among these approaches, computer vision algorithms have proven particularly effective due to their ability to recognize and detect specific morphological details of fish species [7], [8]. Within computer vision–based systems, four primary application areas have been identified: fish detection and recognition algorithms, biomass estimation algorithms, behavior classification systems, and health analysis programs [8].

Contemporary research on fish detection and biomass estimation reveals three prominent trends. First, deep learning algorithms are extensively utilized due to their capability to handle complex data, achieving detection accuracies exceeding 97% in fish counting and relative errors below 3% in weight estimation [5], [6], [8]–[15]. Second, YOLO architectures have been widely adopted for detection, classification, and measurement tasks, appearing in 47% of reviewed articles [6], [8], [16], offering an excellent balance between performance and speed with reported accuracies ranging from 95% to 99% [1], [16]–[21]. Third, transformer-based networks have emerged as powerful tools, particularly for analyzing animal behavior and capturing global dependencies in images [8].

Regarding implementation methodologies, stereo vision-based image processing algorithms enable the application of triangulation geometry to estimate fish depth, reconstruct 3D environments, and measure length while estimating weight [1], [5], [6], [11], [12], [14]–[16]. Alternatively, monocular vision approaches maintain fish at fixed distances, eliminating depth estimation requirements [10], [13], [22]. For morphological measurement, two primary methodologies dominate: segmentation-based approaches utilizing deep learning networks such as Mask R-CNN or deep CNNs to isolate fish bodies and measure length, achieving error rates of 8.7% or lower than 5%, with MAE values of 1.57 g or MRPE values of 3.50% [5], [6], [10], [13], [22], [23]; and keypoint-based methodologies identifying relevant landmarks on fish bodies to measure morphological characteristics, achieving length measurement accuracy of 94%, relative errors of 7%, or mean errors of ±3% [1], [12], [15], [16]. Additional approaches include classical image processing techniques with GrabCut segmentation achieving MRPE values as low as 0.91% [6], and

hybrid algorithms combining deep learning with transformer layers for fish detection and measurement [12].

The complexity of fish measurement systems increases substantially with water turbidity. In controlled environments where water refraction and turbidity are absent, highly specialized preprocessing is unnecessary to achieve high measurement accuracy [6], [10], [13]. Conversely, systems operating in moderate turbidity require light refraction correction, contrast enhancement, and specialized camera calibration [1], [5], [14]–[16], while high-turbidity environments demand multiple preprocessing layers, additional regression models, and robust network architectures [12], [15], [22], [23]. Notably, most existing systems require training neural networks or machine learning algorithms specifically for each environment and fish species, rendering them unable to encompass diverse environments or species. Furthermore, morphologic measurement systems based on computer vision are not widely implemented in commercial fish farms, including recirculating aquaculture systems (RAS), despite the controlled conditions these systems offer that could enhance algorithmic accuracy, thus representing an opportunity for technological development [4].

In this context, this article presents a length measurement and weight estimation system based on image processing and deep learning algorithms, aiming to contribute to automation, control, and monitoring systems for the biological and morphological conditions of fish while completely avoiding capture and handling, thereby significantly reducing the stress associated with traditional measurement procedures. The proposed methodology is designed to remain robust under environmental variations present even in controlled RAS systems and to facilitate future adaptation to different fish species and more diverse environments, thereby expanding the frontiers of aquaculture monitoring technology.

II. Methodology

The proposed system comprises three stages, which can be observed in the figure 1: (i) *preprocessing*, focused on stereo camera calibration and image conditioning; (ii) *processing*, centered on fish detection and anatomical keypoint localization; and (iii) *postprocessing*, where epipolar geometry estimates the 3D position of the keypoints, enabling standard length (SL) measurement and subsequent weight estimation through an allometric model. SL is defined as the distance from the tip of the snout to the beginning of the caudal fin, which reduces algorithm complexity relative to total length.

A. Experimental Setup

The experiment was conducted in a RAS composed of two cylindrical 500 L tanks (127 cm diameter, 75 cm height) with 27 Piaractus brachypomus individuals each, a mechanical filter, and a 250 L bioreactor. A diagram of this image can be seen in the figure 2, located at the National University of Colombia, Bogotá campus. The trial ran for one month (December 16, 2025 – January 15, 2026) under natural lighting only. At the end of the period, total length, standard length, and weight were manually measured from 15 fish using an ichthyometer and a precision scale, following anesthesia-based capture as the reference protocol.

B. Image Acquisition System

A hermetic transparent acrylic housing ($12\times7\times10$ cm^3) was mounted on the tank rim, positioning a synchronized dual-lens ELP stereo camera (4 MP, color CMOS, rolling shutter) 20 cm below the water surface, connected to a Raspberry Pi 4. The two lenses have a fixed 6 cm horizontal baseline. The device transmits data over Wi-Fi for remote acquisition. The complete data acquisition device and its assembly in the RAS system can be seen in the figure 3.

C. Preprocessing

Stereo calibration was performed with MATLAB's Stereo Camera Calibrator using Zhang's method [24] on a 7×5 chessboard pattern (30×30 mm^2 squares), finding the intrinsic and extrinsic parameters of the stereo camera used. From the captured set, 29 image pairs with the greatest angular variation were retained, yielding a mean reprojection error of 0.764 px. Image pairs were subsequently rectified to enforce horizontal epipolar geometry, and histogram equalization was applied to enhance contrast prior to detection.

D. Processing: Detection and Keypoint Localization

For the fish detection stage, the YOLO-Fish v3 deep learning model was employed to identify and localize fish in stereo images. This single-stage convolutional neural network was selected due to its robustness under challenging aquaculture conditions, including illumination changes, reflections, and water turbidity [25]. The pre-trained model was used without additional fine-tuning, applying a confidence threshold of 95% to ensure that only highly reliable detections were under tested conditions. The resulting bounding boxes from the left and right images were independently cropped and used as input for the subsequent processing stage.

In this step, an additional correlation process was not implemented in the identification of the individuals in each image to ensure that the individual identified in both shots is the same, observing that the detections with greater confidence in both images are usually associated with the same individual, so that anomalous measurements are reported as erroneous measurements.

Fish anatomical keypoints were estimated using the HRNet-W32 (High-Resolution Network) architecture implemented through DeepLabCut [26]. This network predicts the positions of the fish mouth and the beginning of the caudal fin while maintaining high-resolution feature representations, enabling accurate anatomical localization [27]. HRNet has demonstrated superior performance in animal pose estimation and fish keypoint detection compared with alternative architectures [28].

The HRNet-W32 model was trained on 147 annotated fish images (95% training and 5% validation) using Google Colab with GPU acceleration. Training was performed for 200 epochs using the AdamW optimizer and data augmentation

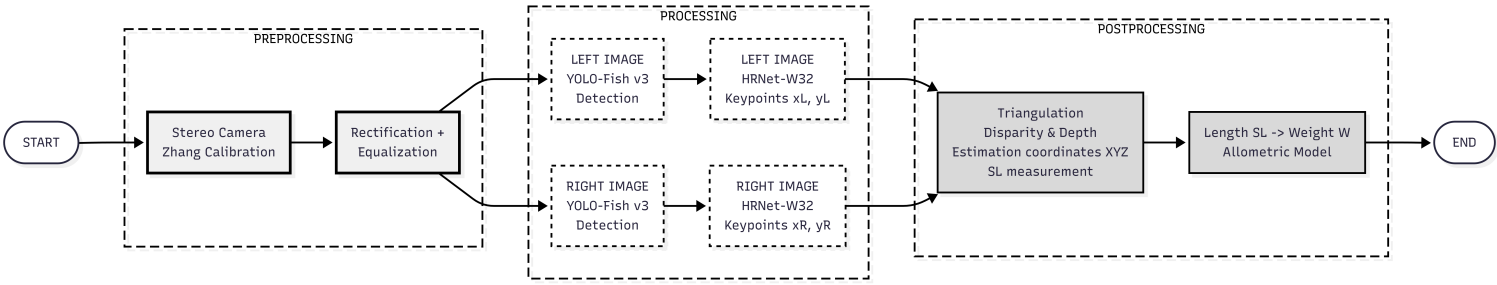


Fig. 1: Complete system architecture.

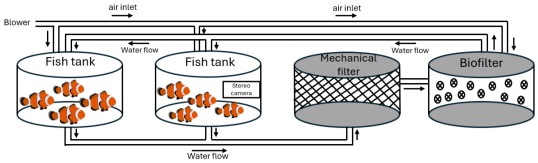


Fig. 2: RAS system diagram.



(a) Hermetic transparent acrylic housing with measuring device.



(b) Implemented system.

Fig. 3: Complete measuring system.

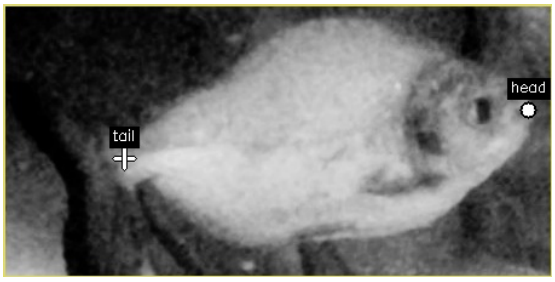


Fig. 4: HRNet-W32 keypoint prediction example showing the detected head (mouth tip) and tail.

techniques, including random rotations, scaling, motion blur, and Gaussian noise. The best-performing model achieved a mean Average Precision (mAP) of 81.30%, a mean Average Recall (mAR) of 86.25%, and a root mean square error (RMSE) of 5.39 pixels, demonstrating reliable localization accuracy for the detected anatomical landmarks under the tested conditions.

Figure 4) shows an example of landmark detection.

E. Postprocessing: 3D Reconstruction and Weight Estimation

Keypoint coordinates from both lenses were mapped back to full-frame coordinates. Since images were rectified, corresponding points share the same image row, and depth Z follows the stereo disparity relation

$$Z = \frac{f \cdot B}{d}, \qquad d = x_L - x_R, \tag{1}$$

where f is the average focal length, B is the stereo baseline, and d is the horizontal , taking into account that x_L y x_R and y are the horizontal coordinates in pixels of the same point in the 2 (left and right) taken by the stereoscopic camera. The full 3D coordinates in the left camera frame are

$$X = \frac{(x_L - c_x) \cdot Z}{f_L}, \qquad Y = \frac{(y_L - c_y) \cdot Z}{f_L}, \tag{2}$$

with (c_x, c_y) the principal point and f_L the left-camera focal length. Standard length is then computed as the 3D Euclidean distance between head and tail keypoints,

$$SL = \sqrt{(X_t - X_h)^2 + (Y_t - Y_h)^2 + (Z_t - Z_h)^2}. \tag{3}$$

Where (X_h, Y_h, Z_h) are the Euclidean coordinates of the head and (X_t, Y_t, Z_t) are the coordinates of the tail.

Next, a statistical analysis is performed comparing the length measurements taken from 15 of the 27 evaluated individuals with the measurements made by the artificial vision system. Since there is no system capable of relating the same fish in both measurement methods, the statistical analysis will be distributional, not paired; that is, a comparison is made between the statistical distribution of both sets of measurements.

Weight is estimated using the allometric length-weight relationship

$$W = a \cdot SL^b, \tag{4}$$

where SL is the standard length in centimeters and a, b are species-specific coefficients. Literature values for *P. brachypomus* ($a = 0.035$, $b = 3$) [29] produced poor performance (MAPE $= 75.66\%$, $R^2 = -7.11$); the model was therefore recalibrated using the 15 manual reference measurements (11–16 cm range), yielding

$$W = 0.0364 \cdot SL^{2.986}. \tag{5}$$

III. Results and Discussion

A. Keypoint Detection Performance

The HRNet-W32 network was trained using DeepLabCut with GPU acceleration on Google Colab. A total of 147 images were manually annotated with two keypoints (head and tail). Training was performed for 200 epochs with a batch size of 8, using the AdamW optimizer with an initial learning rate of 1×10^{-4}, reduced to 1×10^{-5} at epoch 160 and to 1×10^{-6} at epoch 190.Data augmentation included random rotations ($\pm 30°$), scaling (0.5–1.25), motion blur, Gaussian noise ($\sigma = 12.75$), and random 448×448 pixel crop sampling.

The 147 annotated images were split into 139 images (95%) for training and 8 images (5%) for validation, and the resulting model was the one used in the measurement pipeline. The model achieved its best performance at epoch 170, with a mean Average Precision (mAP) of 81.30%, a mean Average Recall (mAR) of 86.25%, and a root mean square error (RMSE) of 5.39 pixels. The tail keypoint showed higher precision (6.51 $\pm$ 3.05 px) than the head keypoint (9.31 $\pm$ 18.26 px), and 87.60% of the predictions fell within 10 pixels of the annotated position (PCK@10px), which corresponds to sub-centimeter localization accuracy. An example of the keypoint localization is shown in Fig. 4.

B. Standard Length Measurement

188 valid measurements were taken using the automated system from the same smaple of 27 individuals, after removing erroneous values from the 196 measurements initially taken, which were compared against the 15 manual measurements. It should be noted that 8 measurements were rejected because the detection of the animal's head and tail keypoints had a confidence level of less than 50 %. Therefore, the reported measurement errors are not directly compared to each other but are determined by the development of the HRNet-W32 artificial intelligence network.

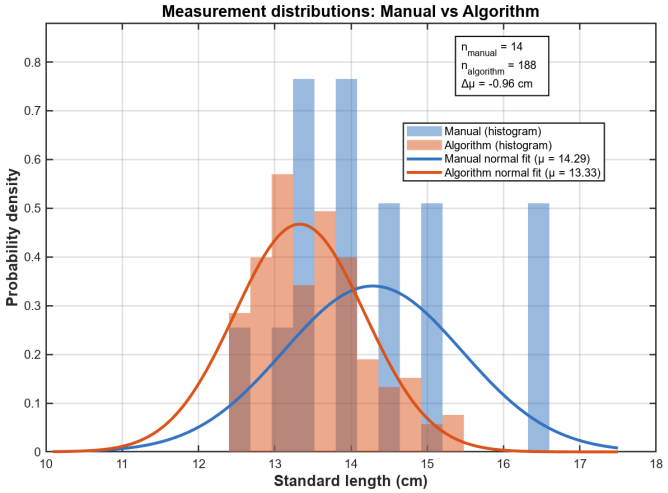


Fig. 5: Overlapping probability distributions of standard length: manual vs. algorithmic measurements.

Furthermore, given the impossibility of matching the manual and automated measurements, a distributional comparison was performed. It is important to note that the 188 measurements were taken from the same 27 individuals, meaning that many of the 188 measurements must be repeated from the same individual, resulting in many of the measurements not being statistically independent. Specifically, there are 2 reasons for the high number of measurements of the automated system: to reach a value close to the average of the measurements of the measured group and to ensure the measurement of all individuals in the system, taking into account that the measured animal does not have a form of direct identification.

Thus, the measurements showed an average of 14.29 ± 1.27 cm (95% CI: [13.64, 14.94]), while automated measurements showed an average of 13.33 ± 0.85 cm (95% CI: [12.99, 13.67]). The Welch test revealed a statistically significant systematic underestimation of 0.96 cm, which can be explained primarily by the keypoint location error of the HRNet network and the lack of a model that considers the interfaces in the light path, in this case, water, acrylic, and air.In summary, there is a 6.7% difference between the manual and automatic measurements, as shown in the image 5. This represents a systematic error and not one directly related to each individual measurement. This could be corrected by designing and attaching a tracking system to each fish to allow for a paired, rather than distributional, study.

C. Weight Estimation

After calibration, the allometric model (Eq. 5) achieved a MAPE of 5.87% and $R^2 = 0.9439$, an order-of-magnitude improvement over the generic-parameter model. Individual errors ranged from 0.27% to 20.13%, with typical errors $\leq$8% within the commercial size range (13-15 cm), corresponding to an absolute deviation of 3–8 g per fish. The Q–Q plot in Fig. 6 confirms close agreement with the identity line; 13 of the 15 evaluated individuals showed errors below 12%. For a

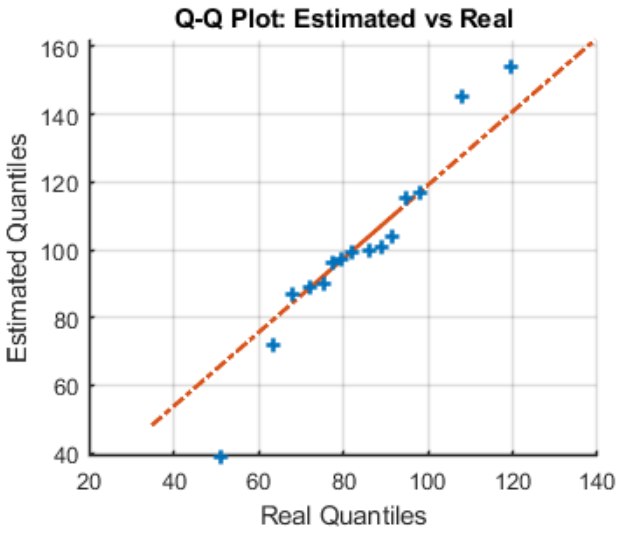


Fig. 6: Quantile–quantile plot comparing estimated vs. measured weights.

100 g fish, the model predicts a value within approximately ± 13 g in 95% of cases. Interestingly, the calibrated weight error is lower than the length error, since parameter fine-tuning implicitly absorbs part of the systematic length bias –the error-propagation chain through the pipeline is therefore not strictly additive.

However, despite the low error percentages reported, the statistical study could only be achieved by adjusting the allometric model parameters using the data collected from the 15 fish measured for the experiment. Therefore, due to the limited experimental data available, this allometric model does not offer generalized validation for the studied species. Furthermore, it should be noted that the study was conducted in an invariant environment with fish that grew under specific conditions without external stimuli that could have affected their growth.

This methodology was developed considering the small amount of data available at the time of the study (15 of the 27 possible measurements) due to logistical and time constraints related to site use. On the other hand, the parameter refinement method yielded good results considering the overall system setup. This method achieved allometric estimation of fish weight using the available data and metrics that validate its performance under the tested conditions. Other methods that strengthen the system's robustness can be considered, either by increasing the model's complexity or by improving the refinement process.Even so, this allometric model represents an advance in the study of non-intrusive weight measurement techniques. Its development could see significant evolution by incorporating measurements from more individuals in more varied environments or by employing more in-depth statistical studies that allow for a better approximation of a generalizable allometric model.

D. Practical Considerations

A specific stereo correspondence algorithm was not implemented. Therefore, the detections were associated based on their confidence levels. This may not guarantee that the same individual is identified in both views, especially when multiple fish are present. An incorrect correspondence can affect the disparity estimation and, consequently, the 3D reconstruction. This limitation should be considered when interpreting the biometric measurements. As future work, a more robust correspondence procedure together with an individual fish tracking system should be implemented to improve the reliability of the measurements.

IV. Conclusions

The current work proposes a non-invasive biometric measurement system for length and weight of Piaractus brachypomus in commercial aquaculture. The system's workflow incorporates the processing of stereo images using YOLOFish for individual detection, and the HRNet-w32 network for detecting markers on the head and tail. Epipolar geometry is then used to measure the standard length of the animal, which is subsequently processed by an allometric model to estimate its weight. This eliminates the need for physical handling of the fish for biometric measurements. The system achieved 87.60 % sub-centimetric keypoint precision (PCK@10px), a length estimation error of 6.7 %, and a weight estimation error (MAPE) of 5.87 % with an R^2 of 0.94. Although systematic length underestimation was observed due to keypoint localization errors and refraction effects, calibration of the allometric model compensated for most of these errors, providing reliable biomass estimation under tested conditions while maintaining greater measurement consistency than traditional manual methods. This development fulfills the objective of implementing a non-intrusive measurement strategy for biometric characteristics in a RAS system. However, logistical and time constraints affected the training of the system's networks, ultimately compromising its robustness for implementation in open environments with uncontrolled conditions and species other than Piaractus brachypomus. Thus, this article demonstrates the potential of this technology through the evaluated system metrics, paving the way for the development of a robust system capable of operating in uncontrolled environments with different species, expanding the training dataset for the neural networks used, and increasing the testing spaces.

Finally, in order to develop a more complete biometrics system, future work proposes a tracking system that can identify the specific individual on whom each of the system's measurements is performed, in-depth experimentation of the system with other species and in other environments, and lastly, adding a refraction model that allows for better work on the interface between water, acrylic, and air when taking photographs.

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